INCH-POUND

A-A-59273A
8 December 2004
SUPERSEDING
A-A-59273
1 November 2004

COMMERCIAL ITEM DESCRIPTION

GENERATOR, SIGNAL
(300 kHz to 2 GHz)

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This Commercial Item Description (CID), describes a Signal Generator in which only those manufacturers meeting or surpassing the following requirements are supplies per this CID. Instruments meeting this CID must be capable of operation within the equipment environment specified herein.

2. SALIENT CHARACTERISTICS. The equipment shall be capable of operation and product sample testing, if so requested, within the accuracies, limits, and specifications herein.

2.1 Classification. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Class 3 in accordance with MIL-PRF-28800, except as specified herein.

2.2 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

2.2.1 Temperature Operating. The equipment shall conform to its specified performance and accuracy while being operated as a volt meter in a temperature range of 0 °C to 50 °C.

2.2.2 Temperature Non-Operating. The equipment shall conform to its specified performance and accuracy after being stored in an environment in the temperature range of -40 °C to 71 °C.

Comments, suggestions, or questions on this document should be addressed to: WR-ALC/LEEC, 295 Byron Street, Warner Robins AFB, GA 31098-1611. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

FSC 6625

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2.2.3 Temperature Non-Operating. The equipment shall conform to its specified performance and accuracy after being stored in an environment in the temperature range of -40 °C to 71 °C.

2.3 Electrical Power Sources. The equipment shall operate from nominal single-phase commercial, military, and shipboard power sources of 110 and 220 volts AC (±10%) at line frequencies of 48 Hz to 63 Hz. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein.

2.4 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 4,000 hours of operation with a statistical certainty of 95%.

2.5 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater, based on an operating time of 2000 hours per year.

2.6 Performance Requirements. The instrument shall meet all requirements specified herein, after the warm-up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions, and operating service condition as specified in MIL-PRF-28800.

2.6.1 Frequency Range. The signal generator shall have a frequency range of 300 kHz to 2 GHz

2.6.1.1 Resolution. The generator shall have a frequency resolution of ≤ 0.1 Hz.

2.6.1.2 Accuracy and Aging Rate. The accuracy of the generator shall be $< 5 \times 10^{-10}$ /day.

2.6.2 Internal Reference Oscillator. The generator shall have a 10 MHz reference frequency.

2.6.2.1 Temperature Stability. The generator shall have a temperature stability of $< \pm 5 \times 10^{-8}$/°C between 0°C to 50°C at a frequency of 10MHz.

2.6.3 Spectral Purity.

2.6.3.1 Harmonics. The generator harmonics shall be ≤ -27 dBc over the entire frequency range when the output < +8dBm.

2.6.3.2 Sub-Harmonics. The sub-harmonics shall be as follows:

500 MHz to 1.2 GHz: ≤ -60 dBc
1.2 GHz to 2 GHz: ≤ -40 dBc

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2.6.3.3 Spurious Non-Harmonics. The spurious non-harmonics, with a 10 kHz offset, shall be as follows:

300 MHz to 1 GHz: ≤ -75 dBc
1 GHz to 2 GHz: ≤ -69 dBc

2.6.3.4 SSB Phase Noise. The generator shall have SSB phase noise of -130 dBc/Hz at 1 GHz with a 20 kHz offset.

2.6.4 RF Output. The output range on this instrument shall be +13 dBm to -127 dBm.

2.6.4.1 Output Resolution. The generator shall have an output resolution of ≥ 0.1 dB.

2.6.4.2 Output Accuracy. The output accuracy shall be ± 1dB.

2.6.4.3 Output Protection. The generator shall have a reverse power protection to 25 Watts.

2.6.5 Amplitude Modulation (AM).

2.6.5.1 AM Depth. The generator's AM depth shall be at least 0 to 99%.

2.6.5.2 AM Resolution. The generator shall have an AM resolution of ≤ 0.1%.

2.6.5.3 AM Rate. The generator shall have an AM rate of at least dc to 10 kHz with a 3dB bandwidth.

2.6.5.4 AM Distortion. At a 1 kHz rate and 50 % depth, the distortion shall be ≤ 5%.

2.6.6 Frequency Modulation (FM).

2.6.6.1 FM Deviation. The generator's FM maximum peak deviation shall be as follows:

500 MHz to 1 GHz: ≤ 1.5 MHz
1 GHz to 2 GHz: ≤ 2.5 MHz

2.6.6.2 FM Resolution. The generator shall have an FM resolution of at least 1 Hz.

2.6.6.3 FM Rate. The generator shall have an FM rate of at least dc to 100 kHz with a 1dB bandwidth or at least dc to 10 MHz with a 3dB bandwidth.

2.6.6.4 FM Distortion. The generator shall have an FM distortion of ≤ 5%.

2.6.7 Pulse Modulation.

2.6.7.1 On/Off Ratio. The on/off ratio for the pulse modulation shall be as follows:

500 MHz to 1.2 GHz: 50dB or better
1.2 GHz to 2 GHz: 80dB or better

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2.6.7.2 Rise and Fall Time. The rise and fall time of the pulse shall be < 100 nanoseconds.

2.6.8 Internal Modulation Source. The generator shall have an internal modulation source for the following waveforms: sine, square, and triangular.

2.6.9 Remote interface. The generator shall come with an IEEE 488.2 standard interface.

2.6.10 Dimensions. The width and depth of the equipment shall be compatible with mounting in an ANSI/EIA-310-D standard equipment rack. The height shall be compatible with four standard height units, 4U. One unit or "U" is defined by ANSI/EIA-310-D to be 1.75 inches or 44.45 mm.

2.6.11 Weight. The total weight of the equipment, excluding accessories and manuals, shall not exceed a two-person lift of 74 pounds, (33.6 kilograms).

2.6.12 Rack Mount. The analyzer shall be capable of being rack-mounted in a standard ANSI/EIA-310-D equipment rack and include a rack-mount conversion kit if needed.

2.7 Manuals. The equipment shall be delivered with operator, maintenance/service, illustrated parts, and calibration manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

If evidence substantiating that the Mean Time Between Failures (MTBF) of the instrument exceeds 20,000 hours is submitted, the maintenance/service manual shall not be required.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. QUALITY ASSURANCE PROVISIONS.

4.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5. PACKAGING.

5.1 Preservation, Packing, and Marking. Preservation, packing, and marking shall be as specified in the contract or order.

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6. NOTES.

6.1 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

6.2 Source of documents.

6.2.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained at http://assist.daps.dla.mil or available from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.2.2 The Code of Federal Regulations (CFR) may be obtained at http://www.gpoaccess.gov/cfr/index.html or available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.2.3 Institute of Electrical and Electronics Engineers (IEEE) standards products may be obtained at http://www.ieee.org or available from IEEE Operations Center. Phone: (800)678-IEEE. Mail: IEEE Operations Center, Sales Office, 445 Hoes Lane, PO Box 1331, Piscataway, NJ, 08855-1331, USA.

6.2.4 ANSI/EIA standards can be obtained at http://www.ansi.org or http://www.eia.org or available from the Electronics Industry Association, Engineering Department, 2001 Pennsylvania Ave., N.W., Washington, D.C., 20006. Phone: 1-800-854-7179 (USA and Canada).

Military Interest

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 84

Agent:
Air Force - 99

(Project 6625-F104)

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